

STRATEGIC ADVICE

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This document contains separate sections for Chapters I, II, and III. You can go directly to the material that interests you by using the following hyperlinks.

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BRIEF STRATEGIC ADVICE FOR CHAPTER I

These strategies are often, but not invariably, helpful. They are intended to supplement your own developing ability to do derivations, not to replace it. They provide fallback suggestions when you are baffled, but do not necessarily lead to the simplest, most direct derivations.

The following general strategies originated in those suggested in the Kalish, Montague, & Mar text. The TerryText versions occur in several different chapters; search for “strategy” or “strategies”. There is a nice summary in Section 4 of Chapter 4.

The strategies are given in the order in which they should be applied, i.e. first consider whether Strategy 1 is applicable, then whether Strategy 2 is applicable, and so forth. Note that as a result of applying one of the latter strategies, an earlier strategy, previously inapplicable, may become applicable. So whenever you don’t know what to do next, always run through the strategies starting at the top.

- 0 **What not to do:** Never, ever, write a show line containing a formula that is already available as a premise or as an antecedent line. (Reminder: an *antecedent line* is an earlier line that is unboxed and does not contain an uncanceled Show.)
 - A It cannot be helpful (since you could always just bring the line down by Repetition).
 - B It will likely lead you on a wild goose chase.
 - C It is a sign of disoriented thinking.
 - D Stop. Think calmly about what you are trying to achieve. Review the strategies below.
- 1 **Most derivations begin with a show line containing the conclusion of the argument.**
 - A You can use the Show Command: “Show Conc”
- 2 **If the conditions for boxing and canceling occur, do it.**
 - A There are three ways to box and cancel: DD, CD, ID.
 - B You can immediately box and cancel any show line (including a conditional or a negation), if you obtain the very formula that appears in the show line. This is a direct derivation, DD.
 - 1) Cite the line number of the formula followed by “DD”, or just add “DD” to the end of the justification for the line on which you obtain the formula.
 - C If the show line is a conditional, you can box and cancel as soon as you obtain the consequent. This is a conditional derivation, CD.
 - 1) Cite the line number of the consequent followed by “CD”, or just add “CD” to the end of the justification for the line on which you obtain the consequent.
 - D For any show line (including a conditional), you can box and cancel whenever you obtain a contradiction (any two lines, one of which is the negation of the other). This is an indirect derivation, ID.

- 1) Cite the line numbers of the two contradictory lines followed by "ID".
- E Be alert for whether any of these three conditions hold. It is easy to miss them.
- F Remember: you must always cancel the *last* of the uncanceled show lines, and the line(s) that justify boxing and canceling must be subordinate to the show line. (You can test for these conditions. Put the cursor in the relevant line(s) and check to see that the show line you wish to cancel appears in reverse video.).
- 3 If a WFF that is needed for boxing and canceling occurs in an antecedent line above the show line you wish to cancel, use Repetition (R) to bring it below the show line, and then box and cancel.**
- 4 To derive a conditional, use conditional derivation**
 - A Assume the antecedent
 - 1) ASS CD
 - B Try to show the consequent
 - 1) You can use the Show Command: "Show Cons"
- 5 To derive anything else, use indirect derivation unless another procedure is immediately obvious.**
 - A ASS ID
- 6 You needn't bother to enter the premises on separate lines because you can cite a premise directly wherever you can cite an antecedent line.**
 - A The reference is PR (PRn will avoid a query when there is more than one premise)
 - B Keep an eye on the premises. Usually, you will need to make *some* use of each premise to complete the argument.
- 7 Whenever MP or MT is applicable, apply them.**
 - A You may need to do some double negating or double unnegating to make MP or MT applicable.
- 8 Once you have begun an indirect derivation, determine whether any of the antecedent lines is the negation of a conditional; if so, enter a show line for that conditional.**
 - A You can use the Show Command: "Show Unneg" followed by the line number of the negated conditional.
 - B To derive the conditional, use conditional derivation (see 4 above).
 - C After completing the derivation, use R to bring down the antecedent line that is the negation of the canceled show line, and box and cancel by ID.
- 9 If a conditional occurs in an antecedent line, and neither MP nor MT can be applied directly, it is often helpful to attempt to show its antecedent (or, alternatively, the negation of its consequent).**
 - A You can use the Show Command: "Show Ant" (or, alternatively, "Show NegCons") followed by the line number of the conditional.
- 10 When using indirect derivation, and none of the above Strategic Hints is applicable, it is sometimes useful to begin attempting to derive the individual sentence letters (or their negations) that appear in antecedent lines.**
 - A Writing "Show ~P" will provide "P" as an assumption for indirect derivation. If you follow this with "Show ~Q", you will then obtain "Q" in the same way. Sooner or later the accumulation of these basic formulas should pay off.

BRIEF STRATEGIC ADVICE FOR CHAPTER II

These strategies are often, but not invariably, helpful. They are intended to supplement your own developing ability to do derivations, not to replace it. They provide fallback suggestions when you are baffled, but do not necessarily lead to the simplest, most direct derivations.

Note that as a result of applying one of the latter strategies, an earlier strategy, previously inapplicable, may become applicable. So whenever you don't know what to do next, always run through the strategies starting at the top.

- 0 **What not to do:** Never try to Show a formula that is already available.
 - A It cannot be helpful (since you could always just bring the line down by Repetition, and any Assumption you are allowed to make will be boxed off when you cancel the Show).
 - B It will likely lead you on a wild goose chase.
 - C It is a sign of disoriented thinking.
 - D **Stop.** Think calmly about what you are trying to achieve. Review the strategies below.

- 1 **Begin by using the Show Command: "Show Conclusion" ("Show Conc").**

HOW TO DEAL WITH SHOW LINES:

- 2 **Every show line is associated with a *strategy*. If you can see for yourself how to box and cancel, do it! Otherwise, follow the strategy:**
 - A **If the show line is a *Conditional*,** use conditional derivation.
 - 1) **ASS CD** to assume the antecedent.
 - 2) Derive the consequent.
 - a Pause for a moment to see if it is obvious how to derive the consequent. If not, use "**Show Consequent**" ("**Show Cons**").
 - 3) Immediately after showing the consequent, cite the now-canceled consequent (which will be on line -1), and use **CD** ("**-1 CD**") to box and cancel. (Remember: line -1 is always the most recent antecedent line. See the discussion of Relative Line References in the Derivation Module Help document.)
 - B **If the show line is a *Negation*,** use indirect derivation.
 - 1) **ASS ID** to assume the unnegation of the show line.
 - 2) Derive a contradiction. (Remember: a *contradiction* consists of two lines, one of which is the negation of the other.)
 - C **If the show line is a *Conjunction*,** derive each conjunct separately.
 - 1) Use "**Show Conjunct**" ("**Show Conj**").
 - 2) Immediately after showing the first conjunct, use "**Show Conjunct**" a second time for the second conjunct.
 - 3) Immediately after showing the second conjuncts, cite the two now-canceled conjuncts (which will be on lines -1 and -2), and use **ADJ** ("**-1 -2 ADJ**"), and then immediately box and cancel by **DD** ("**-1 DD**").
 - D **If the show line is a *Biconditional*,** derive the two conditionals separately and then apply **CB** for a direct derivation.
 - 1) Use "**Show Conditional**" ("**Show Cond**").
 - 2) Immediately after showing the conditional in one direction, use "**Show Conditional**" a second time for the conditional in the other direction.
 - 3) Immediately after showing the second conditional, cite the two now-canceled conditionals (which will be on lines -1 and -2), and use **CB** ("**-1 -2 CB**"), and then immediately box and cancel by **DD** ("**-1 DD**").

- E **If the show line is a *Disjunction***, there are two alternatives:
- 1) Alternative 1: Derive the corresponding conditional and apply CDJ for a direct derivation.
 - a Use “**Show Corresponding Conditional**” (“**Show Corr**”).
 - b Immediately after showing the conditional, cite the now-canceled conditional (which will be on line -1), and use **CDJ** (“-1 CDJ”), and then immediately box and cancel by **DD** (“-1 DD”).
 - 2) Alternative 2: Assume the negation of the disjunction and apply **DM**.
 - a Use **ASS ID**, and then immediately use **DM** (“-1 DM”) to derive a conjunction of negations.
 - b This alternative will generate essentially the same lines as the Alternative 1, with less embedding of subderivations.
 - 3) Don’t forget that it may be possible to derive the disjunction directly by **ADD**.
- F **If the show line is *Atomic* (in Chapter II, a sentence letter)**, do an indirect derivation.
- 1) Use **ASS ID** to assume the negation of the show line.
 - 2) Derive a contradiction.

3 Always look for the possibility of using **CD**, **ID**, or **DD** to box and cancel.

A You may need to use **Repetition** or **Double Negation** before boxing and canceling.

HOW TO MAKE USE OF PREMISES AND ANTECEDENT LINES:

(An *antecedent line* is an earlier line that is unboxed and does not contain an uncanceled “Show”).

4 Apply **MP**, **MT**, **BP**, **BT**, and **MTP** whenever they are applicable.

A You may need to do some double negating, some double unnegating, or some simplifying of conjunctions before the **MP**, **MT**, **BP**, **BT**, or **MTP**.

5 If any of the lines is the negation of a compound formula, there are two alternatives:

A Alternative 1: Apply the appropriate derived rule governing the negation.

- 1) If the line is the *negation of a conditional*, apply **NC**.
- 2) If the line is the *negation of a conjunction*, apply **DM1**.
- 3) If the line is the *negation of a disjunction*, apply **DM2**.
- 4) If the line is the *negation of a biconditional*, apply **NB** or **RT87**.

B Alternative 2: If you have already begun an indirect derivation in some antecedent line, use “**Show Unnegation**” of the negated formula (“**Show Unneg n**”).

Immediately after boxing and canceling a show line introduced by “**Show Unneg n**”, bring the line whose negation has just been shown (which will be line n) beneath the last uncanceled Show (unless it is already there) by using use **R** (“n R”), and then immediately box and cancel by **ID** (“-1 -2 ID”).

6 If any of the lines is a conditional or biconditional (and **MP**, **MT**, or **BP**, **BT** cannot be applied directly), use either “**Show Antecedent**” (“**Show Ant n**”) or “**Show Negation (of the) Consequent**” (“**Show NegCons n**”).

A If the antecedent line is a *biconditional*, the Show command, “**Show Ant n**” will generate a query as to which component you wish to show, or, in the case of “**Show NegCons n**”, which component’s negation you wish to show.

- 1) You can avoid the query by adding “**R**” for the right-hand component or “**L**” for the left-hand component as a parameter. Thus, if line 7 is a biconditional, and you wish to show the negation of the left-hand side, you can use the Show command, “**Show NegCons 7/L**” (Show commands are *not* case-sensitive.)

B Immediately after boxing and canceling a show line introduced by “**Show Ant n**”, cite the relevant conditional or biconditional (which will be on line n), the now-canceled component (which will be on line -1), and apply either **MP** (“n -1 MP”) or **BP** (“n -1 BP”).

- C Immediately after boxing and canceling a show line introduced by **“Show NegCons n”**, cite the relevant conditional or biconditional (which will be on line n), the now-canceled negation of a component (which will be on line -1), and apply either **MT** (**“n -1 MT”**) or **BT** (**“n -1 BT”**).
- 7 **If any of the lines is a disjunction (and MTP cannot be applied directly), use “Show Negation (of a) Disjunct” (“Show NegDisj n”).**
- A Immediately after boxing and canceling a show line introduced by **“Show NegDisj n”**, cite the relevant disjunction (which will be on line n), the now-canceled negation of a disjunct (which will be on line -1), and apply **MTP** (**“n -1 MTP”**).

FINAL FALLBACK STRATEGY:

- 8 **When none of the above suggestions seem helpful, and you have already begun an indirect derivation in some antecedent line, try to derive the individual sentence letters or their negations.**
- A Writing **“Show ~P”** will provide **“P”** as an assumption for indirect derivation. If you follow this with **“Show ~Q”**, you will then obtain **“Q”** in the same way. Sooner or later the accumulation of these basic formulas should pay off.

BRIEF STRATEGIC ADVICE FOR CHAPTER III

The strategic Advice for Chapter III builds upon the Strategic Advice for Chapter II, which will *not* be repeated.

In this Chapter and beyond, it is especially important to remember that these strategies are intended to supplement your own developing ability to do derivations, not to replace it. They provide fallback suggestions when you are baffled, but rarely lead to the simplest, most direct derivations.

The single new issue to arise in Chapter III is the treatment of quantifiers, in particular, the treatment of universal and existential generalizations. The techniques that depend only on the primitive rules are very straightforward.

9 If the show line is a *Universal Generalization*, there are two ways to proceed:

A **Use ASS ID** to assume the negation of the universal generalization and then apply the derived rule **QN1**.

1) This requires that you first succeed in proving **T203**, which requires one **EG** step and one **Universal Derivation**, to enable the rule.

B An alternative strategy is to **Do a universal derivation**

1) Use “Show Instance” (“Show Inst”).

2) Immediately after showing the instance, cite the now-canceled show line (which will be on line -1) and immediately box and cancel by **UD** (“-1 UD”).

10 If the show line is an *Existential Generalization*, there are two ways to proceed:

A **Use ASS ID** to assume the negation of the existential generalization and then apply the derived rule **QN3**.

1) This requires that you first succeed in proving **T204**, which requires one **EG** step and one **Universal Derivation**, to enable the rule.

B An alternative strategy is to **Attempt to derive an instance of the formula that follows the quantifier phrase, and apply the rule EG**.

1) Often, the instance will involve a variable obtained by **EI**.

2) The weakness of this approach as a fallback strategy is that if you do not see how to obtain such an instance, there is no way to predict what instance to try to show.

11 If an antecedent line is an existential generalization, apply EI.

A Since the instantial variable must be entirely new to the derivation, it is generally prudent to begin early in the alphabet, where the variables are little used, and **EI** first to, say, “i”, next to “j”, next to “k”, and so on.

12 If an antecedent line is a universal generalization, apply UI.

A Always complete potential **EI** steps before you **UI**.

1) This is because you can **UI** to a variable that has already been introduced by **EI**, but you cannot **EI** to a variable that has already been introduced by **UI** (or that has been introduced in any other way).

B It is often useful to **UI** to variables that have already been introduced.

1) For example, if you have obtained “ Fi ” and you have “ $\forall x(Fx \rightarrow Gx)$ ” as an antecedent line, you can **UI** to “i” to obtain “ $Fi \rightarrow Gi$ ”, and then apply **MP**. If you were to **UI** to, say, “x” to obtain “ $Fx \rightarrow Gx$ ”, you would not be able to combine this result with “ Fi ” in that way.

13 Free variables in premises or conclusion.

- A When there is a free variables in a premise, it may be advantageous to begin your derivation by deriving a universal closure of the premise. (A *universal closure* of a formula is formed by prefixing a string of universal quantifiers, one for each free variable.)
- 1) If a premise contains free occurrences of just one variable, begin the derivation with a show line to show the universal closure of the premise. Then simply bring in the premise, and box and cancel by UD. If there are free occurrences of many different variables, begin the derivation with a show line to show a universal closure of the premise, and then use “Show Instance” until you are down to the last of the string. Bring in the premise, and box and cancel all the way up by a string of UD steps.
 - 2) Having derived a universal closure of the premise, you can then begin your derivation of the conclusion of the argument.
 - 3) There are valid arguments that seem to require this technique (or a variant), for example, “ $\forall x \rightarrow P. \forall y \wedge Gx \therefore P$ ” (Deriv 3.420) and Deriv 3.422.
- B When there is a free variables in the conclusion, it may be advantageous to begin your derivation by attempting to show a universal closure of the conclusion.
- 1) Follow the show line by “ASS ID QN EI”.
 - 2) If you succeed in showing the universal closure, use “Show Conclusion” and UI to complete a direct derivation of the conclusion.
 - 3) There are valid arguments that seem to require this technique (or a variant), for example, Deriv 3.421 and Deriv 4.421.

A COMPREHENSIVE STRATEGY FOR DOING ANY DERIVATION

As it happens, there is a comprehensive strategy for doing quantificational derivations. The procedure we will describe is a form of quantifier elimination, sometimes referred to as the “EI/UI method”. (The essence of this method is carefully described in section 4 of Chapter IX of the Kalish, Montague, & Mar text, referenced above.) Unfortunately, when followed strictly, this procedure may clutter the derivation with irrelevant lines that can make it increasingly difficult to keep track of what is going on. So you should attempt to combine the main ideas of the EI/UI method with some insight as to where you are going.

The procedure depends on the derived Quantifier Negation rules QN1-QN4, and makes no use of the primitive EG rule and the primitive UD method for boxing and cancelling. It uses only the QN rules plus UI and EI. (Obviously, the primitive EG rule and the primitive UD form of derivation are required to prove the two QN theorems.)

In our version, the method has two parts (which you may need to alternate between). The first part uses the strategies of Chapter II to extract from the premises and conclusion a collection of lines each of which either begins with a quantifier phrase or with the negation of a quantifier phrase. Note that the methods of Chapter II, if followed strictly, always take us ultimately to a show line that is followed by an assumption for Indirect Derivation (**ASS ID**). Thus, if the original argument was valid, a contradiction is introduced into the collection of lines. This is important for the EI/UI method: it is used in a way that presupposes that an implicit contradiction has already been introduced in an antecedent line through an assumption for indirect derivation. The second part is as follows:

- 1 If the show line is a *Universal Generalization*, do an indirect derivation.**
A As in A of 9 above.
- 2 If the show line is an *Existential Generalization*, do an indirect derivation.**
A As in A of 10 above.
- 3 Use QN to convert any line that begins with the negation of a quantifier phrase into one beginning with a quantifier phrase.**
- 4 Begin the following process of EI and UI to eliminate quantifiers**
 - A If the formula begins with an existential quantifier, apply EI.
 - 1) Since the instantial variable must be entirely new to the derivation, it is generally prudent to begin early in the alphabet, where the variables are little used, and EI first to, say, “i”, next to “j”, next to “k”, and so on.
 - B If the formula begins with a universal quantifier, apply UI.
 - 1) When you first apply UI to a formula, you should, according to this strategy, produce a string of UI lines, one for each name letter or variable that occurs free in a line antecedent to the UI line.
 - a Example: If you are applying UI to “ $\forall x\exists y(Fx \rightarrow Gy)$ ”, and the variables “i” and “j” and a name letter all appear free in premises or lines antecedent to the line in which UI is applied, you would UI three times, obtaining the three lines: “ $\exists y(Fi \rightarrow Gy)$ ”, “ $\exists y(Fj \rightarrow Gy)$ ”, and the same for the name letter. (Keep in mind that you need not go through all the tedium of following the standard method if you see a quicker way to do the derivation.)
 - 2) Whenever a new free variable is introduced, perhaps by EI, after you have completed the UI process on a given formula, you should, strictly according to the method, immediately cycle back, and UI the given formula to the new variable.
 - C To the degree possible, try to complete all the EI steps before beginning the UI steps (“EI before you UI!”).

- 1) If it is possible (in a given subderivation) to complete *all* the EI steps before beginning *any* of the UI steps, the process will terminate.
- 2) If it is not possible (in a given subderivation) to complete *all* the EI steps before beginning any of the UI steps, the process will never terminate. For example, if one of the premises were " $\forall x \exists y (Fx \rightarrow Gy)$ ", each UI would lead to an EI that introduced a new free variable. This would require cycling back to UI the premise to the new variable, which would lead to an EI that introduced *another* free variable. This would require cycling back *again* to UI the premise to the new, new variable. And so on *ad infinitum*. Fortunately, the fact that this process does not terminate, does not prevent *the derivation* from terminating. (See step 5 B.)

5 Continue this process, applying QN, EI, and UI until you reach one of the following results.

- A The process terminates, delivering a collection of quantifier-free formulas.
 - 1) In this happy case, the quantifier-free formulas will contain an *implicit* contradiction.
 - 2) You must now extract an *explicit* contradiction from the quantifier-free formulas by using the strategies of Chapter II. The strategies of Chapter III have nothing to add to that process.
 - a Derivation problems 3.001-3.003, 3.008, 3.010, 3.012, 3.016, 3.021, and 3.022, several of which are solved as examples, all terminate without subderivations. You may wish to begin with these simpler problems.
 - b Derivation problems 3.004-3.007, 3.014, 3.015, 3.018, 3.020, and 3.025-3.029, several of which are solved as examples, require subderivations, but each subderivation terminates.
 - c Derivation problems 3.009, 3.011, 3.013, 3.017, 3.019, 3.023, and 3.024, one of which is solved as an example, are somewhat more complex since they involve quantifiers within the scope of other quantifiers, but each subderivation should terminate.
- B It is apparent that the process will not terminate.
 - 1) Even in this situation, the method, if properly applied, will eventually generate enough quantifier-free lines to contain an implicit contradiction.
 - 2) You must be constantly tracking (and perhaps simplifying and extracting results from) the quantifier-free lines as they appear. The process will never terminate, but at some point, the accumulated quantifier-free lines will yield an explicit contradiction using only the strategies of Chapter II.
 - a Derivation problems 3.033-3.036, and 3.039, two of which are solved as examples, are of this kind.
 - b Such problems require more sophisticated derivations. As in the case of all strategies, some problems will yield more quickly to a less strict, more creative, application of the method.
 - 3) When the process doesn't terminate, you must be sure to adopt a method that weaves back and forth among the available quantified lines to be sure that no quantified formula is neglected.
 - a For example, if there were two non-terminating premises, say " $\forall x \exists y (Fx \rightarrow Gy)$ " and " $\forall x \exists y (Gx \rightarrow Fy)$ ", you must be sure not to just keep cycling on one of them and ignore the other. You can ensure against this by entering each of the relevant quantified formulas into any subderivation, and then going down the list, treating each one in turn (EI or UI), being sure to *immediately* cycle back and UI any earlier universal quantifiers, whenever a new variable is introduced.

6 Here is a brief synopsis of the method.

- A Use the methods of Chapter II to extract quantifier-first formulas and negations of quantifier-first formulas from the premises and conclusion.
- B Use QN to transform any negation of a quantifier-first formula into a quantifier-first formula.
- C Apply EI to every existential formula.
- D Apply UI to every universal formula, once for each free variable or name letter that has been introduced.

- E If this process terminates (see above), then the quantifier-free formulas that have been generated will contain an implicit contradiction. Derive an explicit contradiction from them, using the methods of Chapter II.
- F If this process does not terminate, then keep track of the quantifier-free lines, and continue the process until you generate a set of quantifier-free formulas from which, using the methods of Chapter II, you can derive an explicit contradiction.

The following material is *not* strategic advice for doing derivations; it need not be mastered to use the strategies described above. However, some students may find it of interest.

SOME METATHEORETIC CONSIDERATIONS REGARDING THE EI/UI METHOD

Why is it that if the process terminates, the quantifier-free formulas will contain an *implicit* contradiction? Because if the collection of *quantified* formulas contain an implicit contradiction, the quantifier-free formulas must also contain an implicit contradiction. Here is the reason: suppose we take the free variables and name letters of the quantifier-free formulas as each standing for an individual, and as together constituting *all* the individuals of a possible state of affairs. Then, if it were possible for the quantifier-free formulas to all be true they would make all the universal generalizations and existential generalizations true as well, since any universal generalizations would have all of its instances true (because of the requirement to UI to each of the variables and name letters), and each existential generalization would have at least one of its instances true (because of the requirement to EI to at least one variable). Hence if *quantified* formulas contain an implicit contradiction, the quantifier-free formulas must also be implicitly inconsistent. (This argument, like others in this section, requires some more machinery to be made precise.)

Can we predict when the process will terminate? Here are some relevant considerations:

- 1 The process will always terminate when no quantifier phrase lies within the scope of another quantifier phrase. Such formulas are said to have no “overlay” of quantifiers.**
 - A If all premises and the conclusion begin with a single quantifier phrase (or the negation of such), the process will terminate, and nothing other than QN will be required to establish the initial collection of lines to which to apply the EI/UI method.
 - 1) This is the case for derivation problems 3.001-3.003, 3.008, 3.010, 3.012, 3.016, 3.021, and 3.022, several of which are solved as examples. You may wish to begin with these simpler problems.
 - B Even if more than one quantifier phrase occurs in a single premise or in the conclusion, or the quantifier phrase (or its negation) does not *begin* the formula, the EI/UI process will terminate in each subderivation, whenever there is no overlay. However, there may be several sub-derivations. For example, if the conclusion were “ $\forall xFx \wedge \forall xGx$ ”, there would be at least two subderivations, one for each conjunct. Still, if the premises contained no overlay of quantifiers, the process would terminate in each subderivation. Because one must first use the strategies of Chapter II to extract from the premises and conclusion a collection of lines each of which either begins with a quantifier phrase or with the negation of a quantifier phrase, and only then does one begin a subderivation applying the EI/UI method, these derivations are a bit more sophisticated.
 - 1) Derivation problems 3.004-3.007, 3.014, 3.015, 3.018, 3.020, and 3.025-3.029, several of which are solved as examples, are of this kind.
- 2 The process will always terminate if, when in prenex form, a form in which all quantifier phrases in the formula must have scope over all sentential connectives in the formula, no existential quantifier follows a universal quantifier.**

- A Even with overlay, it is always possible with such formulas to complete all the EI steps, if any, before beginning the UI steps. There will then be only a fixed number of UI steps before the process terminates.
- B In a non-prenex formula with overlay, it may not be immediately obvious whether a quantifier in a subformula will appear as existential or universal when it makes it to the front, for application of EI or UI, in a subderivation.

- 1) Example 1: Consider the premise " $\forall x(\exists y Fy \rightarrow Gx)$ ". After applying UI, we get, say, " $(\exists y Fy \rightarrow Gi)$ ". Now if we cannot apply MP or MT, we will start a subderivation as follows:

Show $\exists y Fy$	Show Ant
$\sim \exists y Fy$	ASS ID
$\exists y \sim Fy$	QN.

So the internal existential quantifier becomes universal by the time we are ready to apply the EI/UI process to it.

- 2) Example 2: Consider the premise " $\forall x(Gx \rightarrow \exists y Fy)$ ". After applying UI, we get, say, " $(Gj \rightarrow \exists y Fy)$ ". Now if we cannot apply MP or MT, we will start a subderivation as follows:

Show $\sim \exists y Fy$	Show NegCons
$\exists y Fy$	ASS ID.

in which case, the internal existential quantifier stays existential when we are ready to apply the EI/UI process to it. Note that even if we had used Show Ant, been successful, and applied MP, we would still be left with the existential formula.

- 3) It is possible to compute how a given embedded quantifier will appear when brought to prenex form by using the derived Confinement Laws (see especially T221 and T222).
- 4) Derivation problems 3.009, 3.011, 3.013, 3.017, 3.019, 3.023, and 3.024, one of which is solved as an example, are of this kind. Each of them involves overlay, but each subderivation should terminate.